

Riemannian Manifolds

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December 20, 2025

Abstract

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1 Riemannian Metrics

1.1 The Riemannian Metric

Definition 1.1.1 (Riemannian Metric)

Let M be a smooth manifold. A Riemannian metric on M is a smooth section $g \in \Gamma(M, (TM \otimes TM)^*)$ such that $g_p : T_p(M) \times T_p(M) \rightarrow \mathbb{R}$ is an inner product for all $p \in M$.

Equivalently, a Riemannian metric is an assignment $p \mapsto g_p$ such that for any smooth sections $s, t \in \Gamma(M, TM)$, the map $p \mapsto g_p(s(p), t(p))$ is smooth.

Definition 1.1.2 (Riemannian Manifold)

A Riemannian manifold (M, g) is a manifold M together with a Riemannian metric g on M .

To describe the Riemannian metric in local coordinates, we need some simple algebra. Let V be a finite dimensional vector space over a field k . Let $\{e_1, \dots, e_n\}$ be a basis for V . Let $\{e^1, \dots, e^n\}$ be the dual basis for V^* . Then we may establish a canonical isomorphism

$$(V \otimes_k V)^* \cong V^* \otimes_k V^*$$

given by $f(-, -) \mapsto \sum_{i,j=1}^n f(e_i \otimes e_j) e^i \otimes e^j$. Its converse is given by $\varphi \otimes \psi \mapsto ((v \otimes w) \mapsto \varphi(v)\psi(w))$.

Then isomorphism carries over to vector bundles, and so we have an isomorphism $(TM \otimes TM)^* \cong TM^* \otimes TM^*$. This is how the Riemannian metric is a $(0, 2)$ -tensor field.

Proposition 1.1.3

Let (M, g) be a Riemannian manifold. Let $p \in M$. Let $(U, x^1, \dots, x^n)$ be a chart at p . Then g_p as an element of $T_p(M)^* \otimes T_p(M)^*$ is given locally on the chart by

$$g_p = \sum_{i,j=1}^n g_{i,j}|_p (dx^i)|_p \otimes (dx^j)|_p$$

where $g_{i,j}|_p = g_p \left(\left. \frac{\partial}{\partial x^i} \right|_p, \left. \frac{\partial}{\partial x^j} \right|_p \right)$ where the latter g_p is thought of as an element of $(T_p(M) \otimes T_p(M))^*$.

Theorem 1.1.4

Every smooth manifold admits a Riemannian metric and hence is a Riemannian manifold.

Definition 1.1.5 (Isometries) Let (M, g) and (N, h) be two Riemannian manifolds. We say that (M, g) and (N, h) are isometric if there exists a diffeomorphism $f : M \rightarrow N$ such that

$$h \circ f = g$$

In this case f is said to be an isometry.

Theorem 1.1.6 ((Nash))

Let (M, g) be a Riemannian manifold of dimension n . Then there exists an isometric embedding $f : M \rightarrow \mathbb{R}^{(n+2)(n+3)/2}$ with the standard Euclidean metric.

Definition 1.1.7 (Local Isometries) Let (M, g) and (N, h) be two Riemannian manifolds. We say that they are locally isometric if for all $p \in M$, there exists an open neighbourhood $U \subseteq M$ of p and $V \subseteq N$ open and an isometry $f : U \rightarrow V$.

Definition 1.1.8 (Flat Manifolds) Let (M, g) be a Riemannian manifold. We say that (M, g) is flat if it is locally isometric to $\mathbb{R}^n$ with the standard metric.

In general, not every Riemannian manifold is flat. This can be shown once we discuss curvatures and torsions. However, this is true when $n = 1$.

Lemma 1.1.9 Every 1 dimensional Riemannian manifold is flat.

1.2 Lengths and Angles

Definition 1.2.1 (Length of a Tangent Vector) Let (M, g) be a Riemannian manifold. Let $v \in T_p(M)$ be a tangent vector for $p \in M$. Define the length of v to be

$$|v|_g = \sqrt{g_p(v, v)}$$

Definition 1.2.2 (Angle between two Tangent Vectors) Let (M, g) be a Riemannian manifold. Let $p \in M$. For $v, w \in T_p M$ two tangent vectors, define the angle between v and w to be the unique $\theta \in [0, \pi]$ such that

$$\cos(\theta) = \frac{g_p(v, w)}{|v|_g |w|_g}$$

Definition 1.2.3 (Orthogonal Tangent Vectors) Let (M, g) be a Riemannian manifold. Let $p \in M$. We say that two tangent vectors $v, w \in T_p M$ are orthogonal if

$$g_p(v, w) = 0$$

Definition 1.2.4 (Length of a Curve) Let (M, g) be a Riemannian manifold. Let $\gamma : (a, b) \rightarrow M$ be a curve. Define the length of the curve by

$$L(\gamma) = \int_a^b \sqrt{g_{\gamma(s)}(\gamma'(s), \gamma'(s))} ds$$

Definition 1.2.5 (Angle between two Curves) Let (M, g) be a Riemannian manifold. Let $\gamma_1 : (a, b) \rightarrow M$ and $\gamma_2 : (c, d) \rightarrow M$ be two curves that intersecting at $p = \gamma_1(t_1) = \gamma_2(t_2) \in M$ and that $\gamma_1'(t_1) \neq 0$ and $\gamma_2'(t_2) \neq 0$. Define the angle between γ_1 and γ_2 at p to be the unique $\theta \in [0, \pi]$ such that

$$\cos(\theta) = \frac{g_p(X_{\gamma_1, p}, X_{\gamma_2, p})}{|X_{\gamma_1, p}|_g |X_{\gamma_2, p}|_g}$$

1.3 Musical Isomorphism

Definition 1.3.1 (The Flat Map) Let (M, g) be a Riemannian manifold. Let $p \in M$. For each $X \in T_p M$, define the flat map

$$\flat : T_p M \rightarrow T_p^* M$$

by sending $X \in T_p M$ to the map $X^\flat : T_p M \rightarrow \mathbb{R}$ by $X^\flat(Y) = g_p(X, Y)$.

Theorem 1.3.2 (The Musical Isomorphism) Let (M, g) be a Riemannian manifold. Let $p \in M$. Then the flat map

$$\flat : T_p M \rightarrow T_p^* M$$

is an isomorphism.

Definition 1.3.3 (The Sharp Map) Let (M, g) be a Riemannian manifold. Let $p \in M$. Define the sharp map

$$\sharp : T_p^* M \rightarrow T_p M$$

to be the inverse of the flat map.

1.4 Bundle Metric

Definition 1.4.1 (Bundle Metric) Let M be a topological manifold and $p : E \rightarrow M$ a vector bundle on M . Then a bundle metric on E is a section of $E^* \otimes E^*$ such that it is nondegenerate and symmetric.

In other words, a bundle metric is an assignment to each fibre, an inner product. Bilinearity is seen from $E^* \otimes E^*$, which is exactly the set of all bilinear forms $E \times E \rightarrow \mathbb{R}$.

Proposition 1.4.2 Let M be a smooth manifold. Then a Riemannian metric give rise to a bundle metric on TM . A bundle metric on TM gives rise to a Riemannian metric.

2 Calculus on Manifolds

2.1 Integration of Functions on Riemannian Manifolds

Definition 2.1.1 (The Riemannian Volume Form)

Let (M, g) be an orientable Riemannian manifold. Define the volume form dV of M to be given locally at $p \in M$ on the chart $(U, x^1, \dots, x^n)$ by

$$dV_p = \sqrt{|g_p|} dx^1|_p \wedge \cdots \wedge dx^n|_p$$

2.2 Differential Operators

Definition 2.2.1 (The Gradient of a Function)

Definition 2.2.2 (The Divergence of a Vector Field)

Definition 2.2.3 (The β -Operator)

Definition 2.2.4 (The Curl of a Vector Field)

Proposition 2.2.5

Let (M, g) be an oriented smooth manifold of dimension 3. Then the following diagram is commutative:

$$\begin{array}{ccccccc} \mathcal{C}^\infty(M) & \xrightarrow{\text{grad}} & \mathfrak{X}(M) & \xrightarrow{\text{curl}} & \mathfrak{X}(M) & \xrightarrow{\text{div}} & \mathcal{C}^\infty(M) \\ \text{id} \downarrow & & \downarrow \flat & & \downarrow \beta & & \downarrow \flat \\ \Omega^0(M) & \xrightarrow{d} & \Omega^1(M) & \xrightarrow{d} & \Omega^2(M) & \xrightarrow{d} & \Omega^3(M) \end{array}$$

Definition 2.2.6 (The Laplacian of a Function)

2.3 Calculus on Manifolds

Let M be an oriented smooth manifold of dimension n . let $p \in \partial M$. Then $T_p(\partial M)$ is an $(n - 1)$ -dimensional vector subspace of $T_p(M)$. Recall that a boundary chart of M at $p \in M$ is a chart (U, ϕ) such that the following are true.

- $\phi(U) \subseteq \{(x_1, \dots, x_n) \in \mathbb{R}^n \mid x_n \geq 0\}$.
- $\phi^{-1}(\{(x_1, \dots, x_n) \in \mathbb{R}^n \mid x_n = 0\}) \subseteq \partial M$.
- $\phi(p) = 0$.

Definition 2.3.1 (Unit Normal Vectors)

Let (M, g) be a Riemannian manifold of dimension n with boundary. Let $p \in \partial M$. Define the normal vector $n(p)$ of p to be the unique vector in $T_p(M)$ such that the following are true.

- $n(p)$ is normal to the hyperplane $T_p(\partial M)$ in $T_p(M)$ with respect to the inner product g_p .
- $n(p)$ is a unit vector.
- Choose a boundary chart (U, ϕ) at p . Denote $(\phi_*)_p : T_p(M) \rightarrow T_{\phi(p)}(\mathbb{R}^n)$ the differential of ϕ . Then n th coordinate of $(\phi_*)_p(n_p)$ is less than 0 under the standard basis of $T_{\phi(p)}(\mathbb{R}^n)$.

Lemma 2.3.2

Let (M, g) be a Riemannian manifold of dimension n with boundary. If M is orientable, then the assignment $p \mapsto n(p)$ is a smooth map from ∂M to TM .

Theorem 2.3.3 (The Gradient Theorem)

Let $M \subseteq \mathbb{R}^n$ be a smooth compact connected 1-dimensional manifold. Let $r : [a, b] \rightarrow M$ be a smooth curve such that $\text{im}(r) = M$. Let $f \in \mathcal{C}^\infty(M)$ be a smooth function. Then we have

$$\int_M \nabla f = f(r(b)) - f(r(a))$$

Theorem 2.3.4 (The Divergence Theorem)

Let $M \subseteq \mathbb{R}^n$ be a smooth compact manifold of dimension n with boundary. Let $F : M \rightarrow \mathbb{R}^n$ be a smooth function. Then we have

$$\int_M \text{div}(F) = \int_{\partial M} F \cdot n(x)$$

where both integrals are Riemann integrals.

Theorem 2.3.5

3 The Hodge Star Operator

4 Connections and Parallel Transports

4.1 Affine Connections

Recall that for a smooth vector bundle $p : E \rightarrow M$, we denote the space of smooth sections on E by $\Gamma(E)$. Moreover, $\mathfrak{X}(M)$ is the space of smooth sections on the tangent bundle TM .

Definition 4.1.1 (Connections) Let M be a smooth manifold. Let $p : E \rightarrow M$ be a smooth vector bundle. A connection on p is a map

$$\nabla : \mathfrak{X}(M) \times \Gamma(E) \rightarrow \Gamma(E)$$

where we denote $\nabla(V, T)$ by $\nabla_V(T)$, such that the following are true.

- $C^\infty(M)$ -linearity in first variable: For each $T \in \Gamma(E)$, the map $V \mapsto \nabla_V(T)$ is $C^\infty(M)$ -linear. This means that

$$\nabla_{fV+gW}(T) = f\nabla_V(T) + g\nabla_W(T)$$

for $V, W \in \mathfrak{X}(M)$, $f, g \in C^\infty(M)$.

- $\mathbb{R}$ -linearity in second variable: For each $V \in \mathfrak{X}(M)$, the map $T \mapsto \nabla_V(T)$ is $\mathbb{R}$ -linear. This means that

$$\nabla_V(\lambda T + \mu S) = \lambda \nabla_V(T) + \mu \nabla_V(S)$$

- Product rule: The map ∇ satisfies the following product rule:

$$\nabla_V(fT) = V(f) \cdot T + f\nabla_V(T)$$

Definition 4.1.2 (Affine Connections) Let M be a smooth manifold. An affine connection of M is a connection

$$\nabla : \mathfrak{X}(M) \times \mathfrak{X}(M) \rightarrow \mathfrak{X}(M)$$

on the tangent bundle TM .

The directional derivative is the canonical affine connection on $\mathbb{R}^n$. In fact, every smooth manifold has such a canonical connection that generalizes the directional derivative.

Theorem 4.1.3 Every smooth manifold admits an affine connection.

4.2 Metric Connections

Definition 4.2.1 (Metric Connections) Let M be a smooth manifold. Let ∇ be an affine connection. We say that ∇ is a metric connection if the

$$\nabla(X, g(Y, Z)) = g(\nabla(X, Y), Z) + g(Y, \nabla(X, Z))$$

for all $X, Y, Z \in \mathfrak{X}(M)$.

4.3 The Levi-Civita Connection

Definition 4.3.1 (The Lie Bracket on Smooth Vector Fields) Let M be a smooth manifold. Let $X, Y \in \mathfrak{X}(M)$ be smooth vector fields. Define the Lie bracket of X and Y to be the vector field $[X, Y]$ given by the formula

$$[X, Y]_p(f) = X(Y_p(f)) - Y(X_p(f))$$

for $p \in M$ and $f \in C_{M,p}^\infty$.

Proposition 4.3.2 Let M be a smooth manifold. Let $X, Y \in \mathfrak{X}(M)$ be smooth vector fields. Let

(U, ϕ) be a chart on M . Let X and Y be given locally on the chart by the formula

$$X = \sum_{k=1}^n a_k \frac{\partial}{\partial x^k} \quad \text{and} \quad Y = \sum_{k=1}^n b_k \frac{\partial}{\partial x^k}$$

for $a_k, b_k : U \rightarrow \mathbb{R}$ smooth functions. Then the Lie bracket of X and Y is given locally on the chart by the formula

$$[X, Y] = \sum_{k=1}^n (X(b_k) - Y(a_k)) \frac{\partial}{\partial x^k} = \sum_{i=1}^n \left(\sum_{j=1}^n \left(a_j \frac{\partial b_i}{\partial x^j} - b_j \frac{\partial a_i}{\partial x^j} \right) \right) \frac{\partial}{\partial x^i}$$

Definition 4.3.3 (Levi-Civita Connections) Let (M, g) be a Riemannian manifold. Let $\nabla : \mathfrak{X}(M) \times \mathfrak{X}(M) \rightarrow \mathfrak{X}(M)$ be an affine connection on M . We say that ∇ is a Levi-Civita connection if the following are true.

- ∇ is a metric connection.
- For any $X, Y \in \mathfrak{X}(M)$, we have

$$\nabla(X, Y) - \nabla(Y, X) = [X, Y]$$

Theorem 4.3.4 (Existence and Uniqueness of Levi-Civita Connections) Let (M, g) be a Riemannian manifold. Then M has a unique Levi-Civita connection.

5 Taking Derivatives of the Vector Fields

5.1 Covariant Derivatives

Let M be a smooth manifold. Let $\gamma : I \rightarrow M$ be a smooth curve. Then $\text{im}(\gamma)$ is an embedded submanifold of M . Therefore it makes sense to talk about smooth vector fields on $\text{im}(\gamma)$. We overload the notation and denote the $\mathcal{C}^\infty(\text{im}(\gamma))$ -algebra of smooth vector spaces by

$$\mathfrak{X}(\gamma) = \{X : \text{im}(\gamma) \rightarrow TM \mid X \text{ is a smooth vector field on } \text{im}(\gamma)\}$$

Definition 5.1.1 (Covariant Derivatives Along a Curve) Let M be a smooth manifold. Let $\gamma : (a, b) \rightarrow M$ be a curve on M . Let $\nabla : \mathfrak{X}(M) \times \mathfrak{X}(M) \rightarrow \mathfrak{X}(M)$ be an affine connection on M . The covariant derivative of γ is a map

$$D_t : \mathfrak{X}(\gamma) \rightarrow \mathfrak{X}(\gamma)$$

such that

- $\mathbb{R}$ -linearity: $D_t(aV + bW) = aD_tV + bD_tW$ for $a, b \in \mathbb{R}$.
- Product rule: $D_t(fV) = f'V + fD_tV$ for $f \in C^\infty(a, b)$.
- Extendable: If $V \in \mathfrak{X}(\gamma)$ and there exists $\tilde{V} \in \mathfrak{X}(M)$ such that $\tilde{V}|_{\gamma(t)} = V(t)$ for all $t \in (a, b)$, then $D_tV = \nabla_{\gamma'(t)}\tilde{V}$.

Theorem 5.1.2 (Existence and Uniqueness of Covariant Derivatives) Let M be a smooth manifold. Let ∇ be an affine connection. For each smooth curve $\gamma : (a, b) \rightarrow M$, the connection ∇ determines a unique covariant derivative

$$D_t : \mathfrak{X}(\gamma) \rightarrow \mathfrak{X}(\gamma)$$

For each $t \in I$, choose a chart $(U, \phi = (x^1, \dots, x^n))$ for $\gamma(t) \in M$. Write V locally on the chart by

$$V_{\gamma(t)} = \sum_{i=1}^n a_i(\gamma(t)) \frac{\partial}{\partial x^i} \Big|_{\gamma(t)}$$

Then the unique associated covariant derivatives is given locally on the chart by the formula

$$(D_tV)_{\gamma(t_0)} = \sum_{i=1}^n \left(\frac{\partial a_i}{\partial x^i} \Big|_{\gamma(t_0)} \frac{\partial}{\partial x^i} \Big|_{\gamma(t_0)} + a_i(\gamma(t_0)) \nabla \left(\gamma'(t_0), \frac{\partial}{\partial x^i} \Big|_{\gamma(t_0)} \right) \right)$$

for $t_0 \in I$.

5.2 Parallel Transports

Definition 5.2.1 (Parallel Vector Fields along a Curve) Let M be a smooth manifold. Let $\gamma : I \rightarrow M$ be a curve. Let D_t be its associated covariant derivative. Let $X : M \rightarrow TM$ be a vector field. We say that X is parallel along γ if $D_tX = 0$.

Theorem 5.2.2 Let M be a smooth manifold. Let $\gamma : I \rightarrow M$ be a curve. Let ∇ be an affine connection. Let $t_0 \in I$ and $v_0 \in T_{\gamma(t_0)}M$. Then there exists a unique parallel vector field $V(t)$ along γ such that $V(t_0) = v_0$.

Definition 5.2.3 (Parallel Transports) Let M be a smooth manifold. Let $\gamma : (a, b) \rightarrow M$ be a curve on M . Let $t_0, t \in (a, b)$. The map

$$P_{t_0, t} : T_{\gamma(t_0)}(M) \rightarrow T_{\gamma(t)}(M)$$

defined by $v \mapsto X(t)$ where $X(t)$ is the unique parallel vector field along γ with $X(t_0) = v$.

6 Geometry on Manifolds

6.1 Geodesics

Let M be a smooth manifold. Let $\gamma : I \rightarrow M$ be a smooth curve on M . Then we can compute its differential to obtain a tangent vector

$$\gamma'(t) \in T_{\gamma(t)}M$$

for each $t \in I$. This defines a vector field in $\mathfrak{X}(\gamma)$.

Definition 6.1.1 (Geodesics) Let M be a smooth manifold. Let ∇ be an affine connection on M . A geodesic on M is a curve $\gamma : I \rightarrow M$ such that

$$D_t(\gamma'(t)) = 0$$

where D_t is the associated covariant derivative of ∇ and γ .

We have seen geodesics for metric spaces, but these are slightly different. We require geodesics on manifolds to only minimize the distance locally.

7 Measuring the Curvature

7.1 Gauss-Bonnet Theorem

Theorem 7.1.1 (The Gauss-Bonnet Formula) Let (M, g) be an oriented smooth 2-manifold. Let γ be a positively oriented curved polygon in M and let Ω be its interior. Then

$$\int_{\Omega} K \, dA + \int_{\gamma} \kappa_N \, ds + \sum_{i=1}^k \varepsilon_i = 2\pi$$

where

- K is the Gaussian curvature of g
- dA is the Riemannian volume form
- ε_i are the exterior angles of γ
- The second integral is taken with respect to arc length

Theorem 7.1.2 (Gauss-Bonnet Theorem) Let (M, g) be a smooth compact 2-dimensional Riemannian manifold. Let K be the Gaussian curvature of M and let k_g be the geodesic curvature of ∂M . Then

$$\int_M K \, dA + \int_{\partial M} k_g \, ds = 2\pi\chi(M)$$

Corollary 7.1.3 Let (M, g) be a smooth compact 2-dimensional Riemannian manifold without boundary. Let K be the Gaussian curvature of M . Then

$$\int_M K \, dA = 2\pi\chi(M)$$